Online Appendix: Does Money Buy National Happiness? It Depends on Who You Ask and How

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This Online Appendix complements the paper "Does Money Buy National Happiness? It Depends on Who You Ask and How." References to sections, tables, figures and equations without the prefix "OA" or "A" refer to the main paper. Section OA.1 reproduces the part of the questionnaire of the survey experiment used in the paper, and Section OA.2 presents additional results and robustness checks.

OA.1 Questionnaire of the survey experiment

The survey experiment is described in Section 2.5 of the paper. The survey was programmed in Qualtrics, written in English and translated into the national language(s) of each country. Its main purpose was to measure attitudes toward global policies, which are analyzed in a separate paper (Fabre, 2025); the full questionnaire, in all languages, is available in the replication files of that paper (Fabre, 2025). The survey was approved by the CIRED institutional review board (IRB-CIRED-2025-2) and pre-registered on OSF (https://osf.io/7mzn4).

OA.1.1 Structure of the questionnaire

The questionnaire comprises, in this order: a welcome page; socio-demographic questions (used for the quotas and the weights: gender, age, income, education, region, urbanicity, as well as race in the U.S. and citizenship in Saudi Arabia) and voting; an open-ended question; questions on global policies (a conjoint analysis, global taxes and their revenue use, global climate policies, a global wealth tax, global redistribution, and a tool

to choose one's preferred global redistribution); the *well-being* block; comprehension questions (asked to a subsample); synthetic questions on global policies; and feedback on the survey. The well-being block displays a single question. Each respondent is randomly assigned, with equal probabilities, to one of the four variants below. No other question of the survey varies with the variant assigned, and no other condition of the survey is used in this paper.

OA.1.2 The four variants of the life-evaluation question

1. *Ladder 0–10 (Gallup's original question).* "Please imagine a ladder, with steps numbered from 0 at the bottom to 10 at the top. The top of the ladder represents the best possible life for you and the bottom of the ladder represents the worst possible life for you. On which step of the ladder would you say you personally feel you stand at this time?" Answers: 0 (*Worst possible*), 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 (*Best possible*).

2. *Ladder 1–10.* "Please imagine a ladder, with steps numbered from 1 at the bottom to 10 at the top. The top of the ladder represents the best possible life for you and the bottom of the ladder represents the worst possible life for you. On which step of the ladder would you say you personally feel you stand at this time?" Answers: 1 (*Worst possible*), 2, 3, 4, 5, 6, 7, 8, 9, 10 (*Best possible*).

3. *Satisfaction 0–10.* "All things considered, how satisfied are you with your life as a whole these days?" Answers: 0 (*Completely dissatisfied*), 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 (*Completely satisfied*).

4. *Satisfaction 1–10 (WVS's original question).* "All things considered, how satisfied are you with your life as a whole these days?" Answers: 1 (*Completely dissatisfied*), 2, 3, 4, 5, 6, 7, 8, 9, 10 (*Completely satisfied*).

OA.2 Additional results

	Happy	Very Happy	Very Unhappy	V. Happy – V. Unhappy	Happiness (mean)	Satisfaction (mean)	Satisfied	Happy + Satisfied
Happy	1.00	0.60	−0.73	0.70	0.89	0.73	0.72	0.90
Very Happy	0.60	1.00	−0.37	0.98	0.89	0.60	0.51	0.59
Very Unhappy	−0.73	−0.37	1.00	−0.54	−0.67	−0.58	−0.57	−0.69
V. Happy – V. Unhappy	0.70	0.98	−0.54	1.00	0.94	0.66	0.58	0.67
Happiness (mean)	0.89	0.89	−0.67	0.94	1.00	0.75	0.69	0.83
Satisfaction (mean)	0.73	0.60	−0.58	0.66	0.75	1.00	0.96	0.93
Satisfied	0.72	0.51	−0.57	0.58	0.69	0.96	1.00	0.95
Happy + Satisfied	0.90	0.59	−0.69	0.67	0.83	0.93	0.95	1.00

Table A1: Correlation between national well-being indicators (WVS country-years).
(Back to Section 2.1.)

Panel A: WVS/EVS, 1981–2023

Dependent variable:	Happy	Very Happy	Very Unhappy	V. Happy – V. Unhappy	Happiness (mean)	Satisfaction (mean)	Satisfied	Happy + Satisfied
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\log_{10}$ GDP p.c. (PPP)	0.109***	0.002	−0.022***	0.023	0.264**	1.143***	0.236***	0.173***
	(0.019)	(0.025)	(0.004)	(0.026)	(0.083)	(0.142)	(0.024)	(0.020)
Observations	340	340	340	340	340	340	340	338
R^2	0.144	0.000	0.072	0.004	0.044	0.233	0.316	0.275

Panel B: Gallup World Poll, 2006–2023

Dependent variable:	Satisfied	Very satisfied (8–10)	Extremely satisfied (9–10)	Completely satisfied (10)	Unsatisfied (0/1–4)	Dissatisfied (0/1–2)	Satisfaction (mean)
	(9)	(10)	(11)	(12)	(13)	(14)	(15)
$\log_{10}$ GDP p.c. (PPP)	0.372***	0.218***	0.065***	0.012**	−0.304***	−0.135***	1.779***
	(0.017)	(0.018)	(0.009)	(0.005)	(0.012)	(0.009)	(0.083)
Observations	2,344	2,344	2,344	2,344	2,344	2,344	2,344
R^2	0.655	0.442	0.178	0.015	0.639	0.432	0.611

Table A2: Slope of national well-being indicators with respect to $\log_{10}$ GDP per capita (PPP).
Note: Each column is an OLS regression across country-years of a national well-being indicator on $\log_{10}$ GDP per capita (PPP, constant 2021 dollars), with a constant. Standard errors clustered by country in parentheses. * $p < .05$, ** $p < .01$, *** $p < .001$.
(Back to Section 3.2.)

Data	Indicator	Obs.	log GDP PPP	Best income variable	Region	log GDP PPP + Region
WVS, all waves	Happy	340	0.12	0.18	0.26	0.32
WVS, all waves	Very Happy	340	−0.03	−0.04	0.27	0.27
WVS, all waves	Very Unhappy	340	0.06	0.10	0.10	0.11
WVS, all waves	V. Happy – V. Unhappy	340	−0.02	−0.02	0.27	0.26
WVS, all waves	Happiness (mean)	340	0.02	0.05	0.29	0.30
WVS, all waves	Satisfaction (mean)	340	0.21	0.23	0.37	0.46
WVS, all waves	Satisfied	340	0.30	0.34	0.36	0.50
WVS, all waves	Happy + Satisfied	338	0.26	0.31	0.35	0.47
WVS, last obs. per country	Happy	115	0.05	0.03	0.11	0.12
WVS, last obs. per country	Very Happy	115	0.01	−0.05	0.17	0.16
WVS, last obs. per country	Very Unhappy	115	0.04	0.06	0.01	0.02
WVS, last obs. per country	V. Happy – V. Unhappy	115	−0.01	−0.08	0.16	0.15
WVS, last obs. per country	Happiness (mean)	115	−0.04	−0.09	0.12	0.10
WVS, last obs. per country	Satisfaction (mean)	115	0.18	0.19	0.30	0.33
WVS, last obs. per country	Satisfied	115	0.29	0.31	0.31	0.38
WVS, last obs. per country	Happy + Satisfied	115	0.24	0.25	0.28	0.34
Gallup, all years (2006–23)	Satisfaction (mean)	2344	0.61	0.66	0.49	0.68
Gallup, all years (2006–23)	Satisfied	2344	0.65	0.72	0.52	0.72
Gallup, all years (2006–23)	Unsatisfied (0/1–4)	2344	0.63	0.65	0.48	0.68
Gallup, last obs. per country	Satisfaction (mean)	167	0.60	0.58	0.45	0.64
Gallup, last obs. per country	Satisfied	167	0.65	0.68	0.52	0.70
Gallup, last obs. per country	Unsatisfied (0/1–4)	167	0.65	0.60	0.48	0.67
WHR, 2023–25 average	Satisfaction (mean)	147	0.62	0.60	0.52	0.69

Table A3: Out-of-sample predictive capacity: leave-one-country-out cross-validated R^2.
Note: Each country's observations are predicted with a model estimated on all other countries. Cross-validated $R^2 = 1 - \sum(y - \hat{y}_{\text{out-of-sample}})^2 / \sum(y - \bar{y})^2$; it can be negative when a model predicts worse than the overall mean.
(Back to Section 3.3.)

Specification	Obs.	Countries	R^2 income		R^2 region	Share of explained variance due to income			Region better
			log PPP	best		log PPP	best	adj. R^2	(share of cases)
WVS, all waves	340	112	0.14	0.20	0.33	0.23	0.33	0.27	0.98
WVS, waves 1–2 (1981–91)	26	21	0.07	0.43	0.58	0.09	0.40	0.10	1.00
WVS, wave 3 (1995–99)	55	55	0.16	0.33	0.65	0.14	0.28	0.18	1.00
WVS, wave 4 (1999–2004)	40	40	0.19	0.31	0.33	0.27	0.47	0.47	0.43
WVS, wave 5 (2004–09)	58	58	0.21	0.35	0.49	0.22	0.38	0.23	0.98
WVS, wave 6 (2010–16)	60	60	0.08	0.22	0.27	0.14	0.41	0.19	0.96
WVS, wave 7 (2017–23)	102	92	0.09	0.17	0.28	0.21	0.35	0.27	1.00
WVS, wave 7 w/o 2020–21	80	72	0.08	0.19	0.29	0.19	0.38	0.27	0.95
WVS, population-weighted	340	112	0.10	0.19	0.21	0.25	0.44	0.32	0.77
WVS, last obs. per country	115	112	0.13	0.19	0.25	0.26	0.39	0.30	0.88
WVS, no GDP imputation	321	108	0.13	0.20	0.32	0.22	0.33	0.27	0.98
WVS, GDP PPP 2017 $ (old)	340	112	0.14	0.20	0.33	0.24	0.33	0.28	0.98
WVS, without EVS surveys	304	108	0.12	0.19	0.37	0.18	0.28	0.22	1.00
Gallup, all years (2006–23)	2344	167	0.42	0.48	0.42	0.46	0.51	0.49	0.31
Gallup, last obs. per country	167	167	0.42	0.44	0.41	0.44	0.48	0.45	0.37
Gallup, last obs., WVS countries	109	109	0.44	0.47	0.43	0.46	0.49	0.47	0.35
Gallup, last obs., 6 regions	167	167	0.42	0.44	0.42	0.43	0.46	0.44	0.41
Gallup, last obs., WB regions	167	167	0.42	0.44	0.38	0.47	0.50	0.48	0.29
WHR, 2023–25 average	147	147	0.63	0.63	0.55	0.56	0.56	0.56	0.00

Table A4: Robustness: income vs. region across specifications and datasets.

Note: Averages over well-being indicators (8 for the WVS; 7 satisfaction indicators for Gallup; mean ladder for the WHR). "Best" refers to the income measure with the highest average R^2 in the main WVS specification (Cluster k=7 nominal). "Region better" is the share of indicator $\times$ income-measure combinations for which $s < 0.5$. Alternative region classifications and exclusions of regions: see Table 3. (Back to Section 3.3.)

Indicator	Other variable	Obs.	R^2 other alone	Shapley decomposition of R^2			
				Income	Region	Other	Total
Satisfaction (mean)	Median income (PIP, log)	339	0.19	0.11	0.32	0.07	0.50
Satisfaction (mean)	Household consumption p.c. (log)	292	0.30	0.10	0.29	0.13	0.51
Satisfaction (mean)	Freedom of choice	334	0.63	0.11	0.21	0.42	0.74
Satisfaction (mean)	Importance of God	329	0.01	0.17	0.33	0.02	0.51
Satisfaction (mean)	Tolerance (homosexuality)	326	0.24	0.12	0.30	0.10	0.52
Satisfaction (mean)	Importance of democracy	216	0.08	0.13	0.32	0.02	0.47
Satisfaction (mean)	GDP growth (5-year)	334	0.02	0.16	0.34	0.02	0.52
Happiness (mean)	Median income (PIP, log)	339	0.03	0.03	0.32	0.01	0.36
Happiness (mean)	Household consumption p.c. (log)	292	0.08	0.03	0.28	0.04	0.35
Happiness (mean)	Freedom of choice	333	0.39	0.02	0.22	0.27	0.51
Happiness (mean)	Importance of God	330	0.01	0.05	0.31	0.03	0.39
Happiness (mean)	Tolerance (homosexuality)	325	0.09	0.02	0.31	0.07	0.40
Happiness (mean)	Importance of democracy	216	0.03	0.02	0.28	−0.03	0.27
Happiness (mean)	GDP growth (5-year)	334	0.01	0.03	0.31	0.00	0.34

Table A5: Other country-level correlates of national well-being: Shapley decomposition of R^2 (WVS).

Note: Log of median daily income or consumption per capita (World Bank PIP line-up estimates) and of household final consumption expenditure per capita (constant 2015 dollars). Country-year averages of WVS variables: perceived freedom of choice and control (1–10), importance of God in one's life (1–10), justifiability of homosexuality (1–10), importance of living in a democracy (1–10, waves 5–7), and average annual growth of real GDP per capita over the previous five years. Shapley decomposition of the R^2 of the regression on log GDP per capita, region dummies and the other variable. (Back to Section 3.6.)

Dependent variable:	Happy	Very Happy	Very Unhappy	V. Happy – V. Unhappy	Happiness (mean)	Satisfaction (mean)	Satisfied	Happy + Satisfied
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\log_{10}$ GDP p.c. (PPP)	0.286^{***}	0.207^{***}	-0.039^{***}	0.246^{***}	1.063^{***}	2.313^{***}	0.440^{***}	0.366^{***}
	(0.043)	(0.048)	(0.010)	(0.054)	(0.180)	(0.420)	(0.076)	(0.058)
Observations	312	312	312	312	312	312	312	310
Within R^2	0.281	0.146	0.044	0.160	0.245	0.344	0.347	0.389

Table A6: Within-country relation between national well-being and $\log_{10}$ GDP per capita (country fixed effects, WVS).

Note: Each column is an OLS regression of a national well-being indicator on $\log_{10}$ GDP per capita (PPP) with country fixed effects, over the country-years of countries surveyed at least twice. Standard errors clustered by country in parentheses. * $p < .05$, ** $p < .01$, *** $p < .001$. On average over the eight indicators, the within-country R^2 is 0.24. (Back to Section 3.6.)

Country	Occurrences as happiest	Region
Switzerland	10	Western
Mexico	9	Latin America
Kyrgyzstan	6	Asia
Australia	5	Western
Netherlands	4	Western
Nigeria	4	Africa
Puerto Rico	4	Latin America
Colombia	3	Latin America
Finland	3	Western
Vietnam	3	Asia

Table A7: Countries most often ranked happiest (WVS, 8 indicators $\times$ 8 samples: each wave and all waves combined).

Note: Looking at all waves combined, the happiest country-year is: Kyrgyzstan–2020 (3 indicators: Very Happy, V. Happy – V. Unhappy, Happiness (mean)); Finland–1981 (Satisfied); Malaysia–2012 (Very Unhappy); Mexico–2012 (Satisfaction (mean)); Qatar–2010 (Happy); Vietnam–2020 (Happy + Satisfied). For *Very Unhappy*, the happiest country-year is the one with the lowest share. (Back to Section 3.7.)

	Answer		Answer (0–10 equiv.)	Satisfied (6+)	Answer (0–10 equiv.)	
	(1)	(2)	(3)	(4)	(5)	(6)
Ladder wording (Gallup)	−0.451***	−0.455***	−0.456***	−0.088***	−0.464***	−0.494**
	(0.046)	(0.067)	(0.067)	(0.013)	(0.123)	(0.172)
Scale 1–10	0.137**	0.132	−0.260***	−0.008	−0.276***	−0.540**
	(0.046)	(0.068)	(0.072)	(0.013)	(0.046)	(0.182)
Ladder wording (Gallup) × Scale 1–10		0.008	−0.041	0.028		0.055
		(0.092)	(0.097)	(0.019)		(0.247)
Income quartile (1–4)					0.689***	0.635***
					(0.031)	(0.043)
Income quartile (1–4) × Ladder wording (Gallup)					−0.004	0.009
					(0.042)	(0.058)
Income quartile (1–4) × Scale 1–10						0.106
						(0.061)
Income quartile (1–4) × Ladder wording (Gallup) × Scale 1–10						−0.022
						(0.083)
Observations	11,000	11,000	11,000	11,000	11,000	11,000
R^2	0.057	0.057	0.059	0.042	0.160	0.160
Country fixed effects	✓	✓	✓	✓	✓	✓

Table A8: Effects of question wording and scale on life evaluation: survey experiment (Fabre (2025)).

Note: OLS with country fixed effects and survey weights (normalized within country). *Ladder wording*: Cantril ladder (vs. WVS life satisfaction). *Scale 1–10*: answers from 1 to 10 (vs. 0 to 10). In columns 3, 5 and 6, 1–10 answers are linearly stretched to 0–10. Columns 5 and 6 interact the treatments with the respondent's income quartile within their country; in column 6, the interaction between income and wording is 0.009 ($p = .883$) and the triple interaction is −0.023 ($p = .786$). Heteroskedasticity-robust standard errors in parentheses. * $p < .05$, ** $p < .01$, *** $p < .001$. (Back to Section 4.3.)

Country	Year WVS/EVS (Gallup)	Survey: mode	Gallup (1)	WVS/EVS (2)	Gap D (2)−(1)	Ladder 0–10 (3)	Satisf. 1–10 (4)	Question effect Q (4)−(3)	Residual R $D-Q$
Poland	2017	EVS: CAPI	6.28	7.56	1.28	5.53	6.37	0.84	0.44
Spain	2017	EVS: CAPI	6.47	7.49	1.02	5.61	6.99	1.38	−0.36
Japan	2019	WVS: Mail	5.94	6.76	0.82	5.33	5.00	−0.33	1.15
Italy	2018	EVS: CAPI	6.69	7.28	0.59	5.99	6.59	0.60	−0.01
Germany	2018	WVS: CAPI	7.15	7.74	0.58	6.01	6.84	0.83	−0.25
United Kingdom	2022	WVS: CAPI/Web/Mail	6.76	7.32	0.56	5.67	6.77	1.10	−0.55
France	2018	EVS: CAPI	6.79	7.26	0.47	5.96	6.68	0.72	−0.25
Switzerland	2017	EVS: Web/Mail/CAPI	7.58	8.02	0.43	6.85	7.31	0.46	−0.03
Saudi Arabia	2003 (2006)	WVS: Face-to-face	7.06	7.28	0.23	6.31	7.10	0.79	−0.57
United States	2017	WVS: Web/Phone	7.23	7.22	−0.01	6.03	6.74	0.70	−0.71

Table A9: Decomposition of the WVS–Gallup gap by country.

Note: Mean answers on native scales (Gallup and ladder: 0–10; WVS/EVS and satisfaction: 1–10); columns (3) and (4): Fabre (2025). The Gallup year is given in parentheses when it differs from the WVS year. Survey: mode: source of the WVS/EVS data and main interview mode(s) (CAPI: computer-assisted personal interviews). (Back to Section 4.3.)

Statistic	Estimate	95% bootstrap CI
Mean gap D (WVS − Gallup)	0.60	[0.55; 0.64]
Mean question effect Q (satisfaction 1–10 − ladder 0–10)	0.71	[0.56; 0.84]
Mean residual $R = D - Q$ (samples, mode, context)	−0.11	[−0.26; 0.04]
Share of mean gap due to question ($\bar{Q}/\bar{D}$)	1.19	[0.94; 1.45]
Share of gross level difference due to question ($\lvert\bar{Q}\rvert/(\lvert\bar{Q}\rvert + \lvert\bar{R}\rvert)$)	0.86	[0.76; 0.99]
Share of cross-country variance of D due to Q ($\mathrm{cov}(D,Q)/\mathrm{var}(D)$)	0.11	[−0.33; 0.55]
Correlation between D and Q	0.09	[−0.26; 0.41]
Mean absolute gap $\lvert D\rvert$	0.60	[0.56; 0.65]
Mean absolute residual $\lvert R\rvert$	0.43	[0.33; 0.60]
Mean gap, share satisfied (6+)	0.07	[0.06; 0.08]
Mean question effect, share satisfied (6+)	0.11	[0.08; 0.14]
Variance share due to Q, share satisfied	0.04	[−0.33; 0.45]

Table A10: Decomposition of the WVS–Gallup gap: summary.

Note: 10 countries. 95% confidence intervals from 1,000 bootstrap replications resampling respondents within country × variant (new survey) and within country (WVS). For Gallup, only the distribution of answers is available: in each country, the numbers of respondents giving each answer (0 to 10) are redrawn from a multinomial distribution with the observed shares and sample size, which is equivalent to resampling respondents. (Back to Section 4.3.)

Measure	Surveys	Median				Correlation
		All	GDP < 10k	10–30k	> 30k	with log GDP
Tertiary education, 25+ (ratio sample/benchmark)	295	1.18	1.98	1.18	1.05	−0.40
Urban (ratio sample/benchmark)	76	1.44	2.19	1.45	1.22	−0.69
Aged 65+ (ratio sample/benchmark)	335	0.90	0.64	0.88	1.01	0.33
Women among married respondents − 0.5	337	0.01	−0.00	0.01	0.01	0.19

Table A11: Representativeness of WVS/EVS samples: sample shares relative to population benchmarks.

Note: One observation per WVS or EVS survey. Ratios: weighted sample share divided by the population share, medians across surveys (a ratio above 1 indicates over-representation). Benchmarks (World Bank WDI): share of the population aged 25+ with at least short-cycle tertiary education (UNESCO), or with complete or incomplete tertiary education (Barro & Lee, 2013) before 2011; urban population share (national definitions; sample: urban and peri-urban respondents, waves 4, 5 and 7 only); share aged 65+ among the population aged 15+ (sample: 65+ among respondents, usually 18+). Women among married: unweighted share of women among married or cohabiting respondents minus 0.5. Correlations with log GDP per capita (PPP) of the log ratios (or of the deviation). (Back to Section 4.4.)

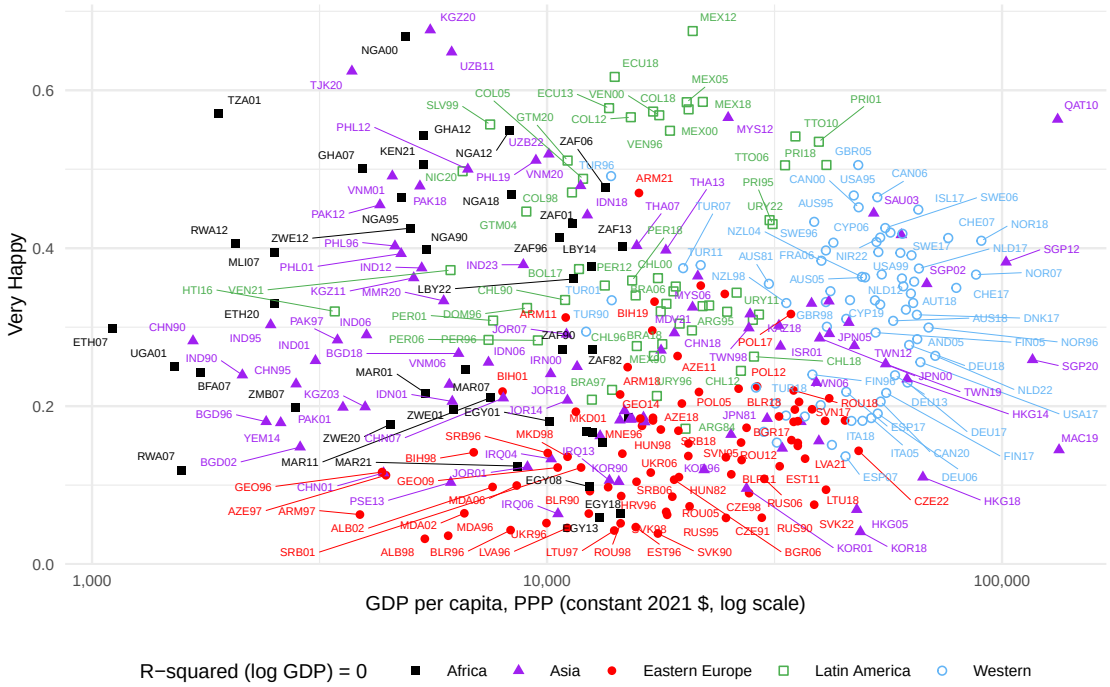


Figure A1: Share of *Very happy* people vs. GDP per capita (PPP), WVS/EVS 1981–2023.

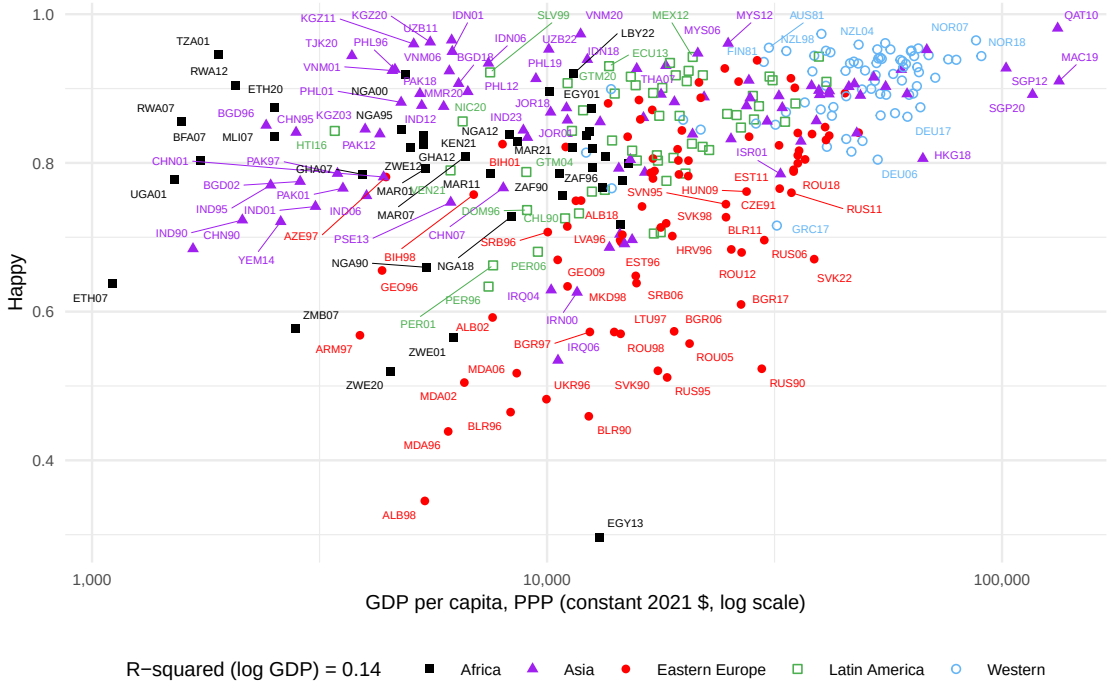


Figure A2: *Happy* vs. GDP per capita (PPP), WVS/EVS 1981–2023.

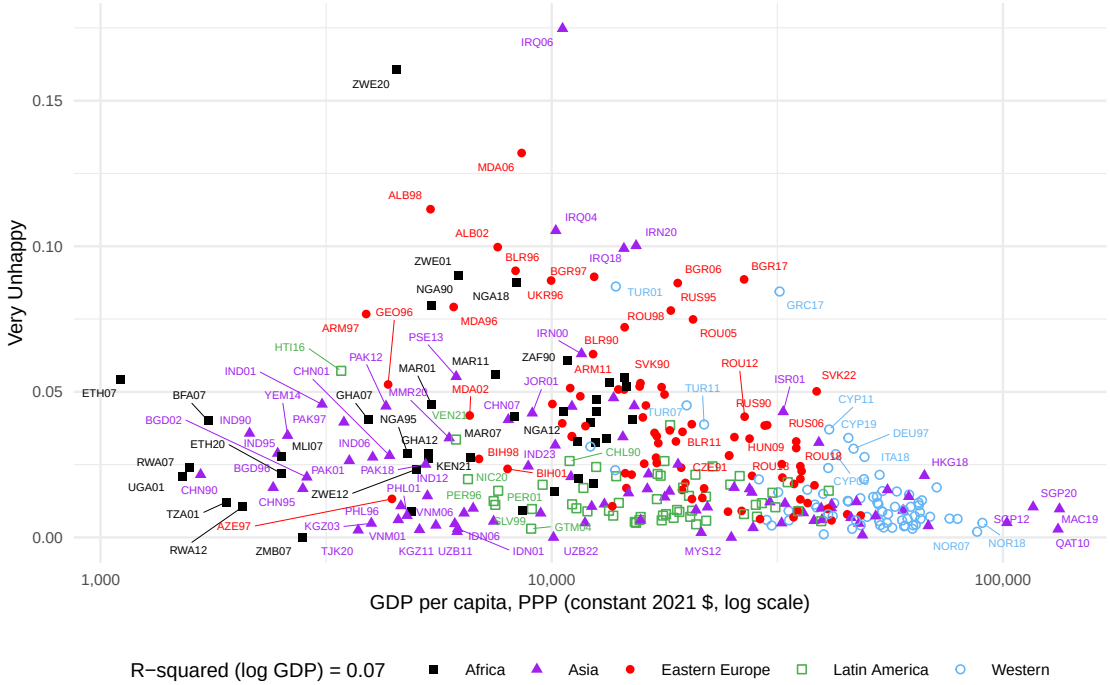


Figure A3: *Very Unhappy* vs. GDP per capita (PPP), WVS/EVS 1981–2023.
Note: Egypt 2013 (42% of very unhappy people) is not shown, to keep the y-axis readable; the R^2 in the legend includes it.

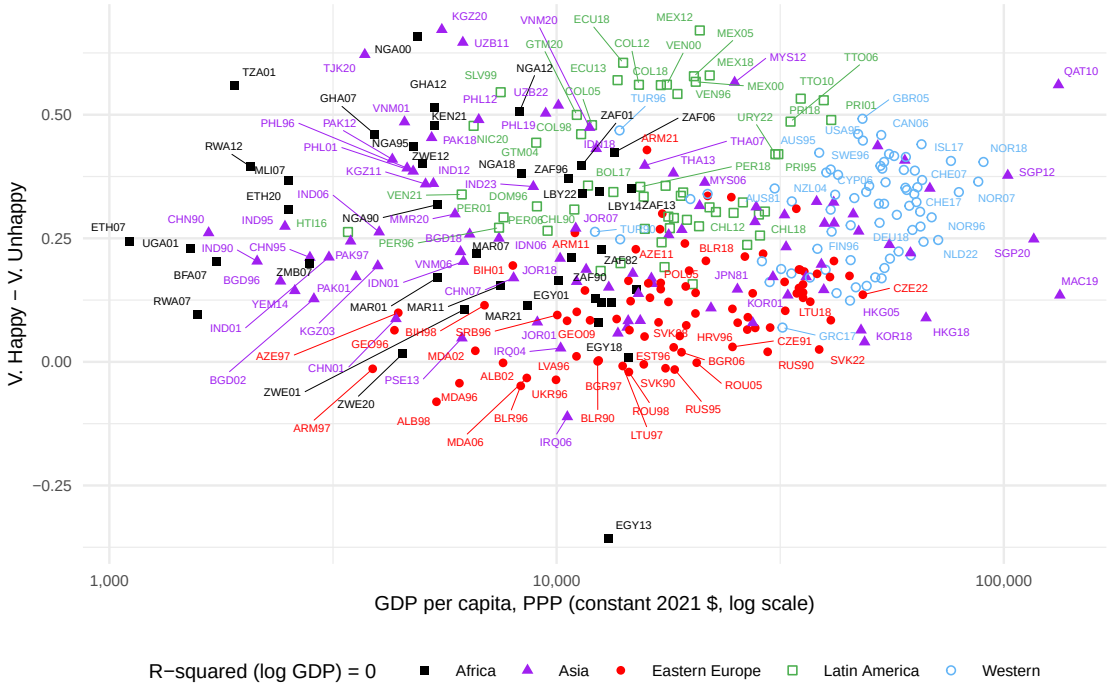


Figure A4: *V. Happy – V. Unhappy* vs. GDP per capita (PPP), WVS/EVS 1981–2023.

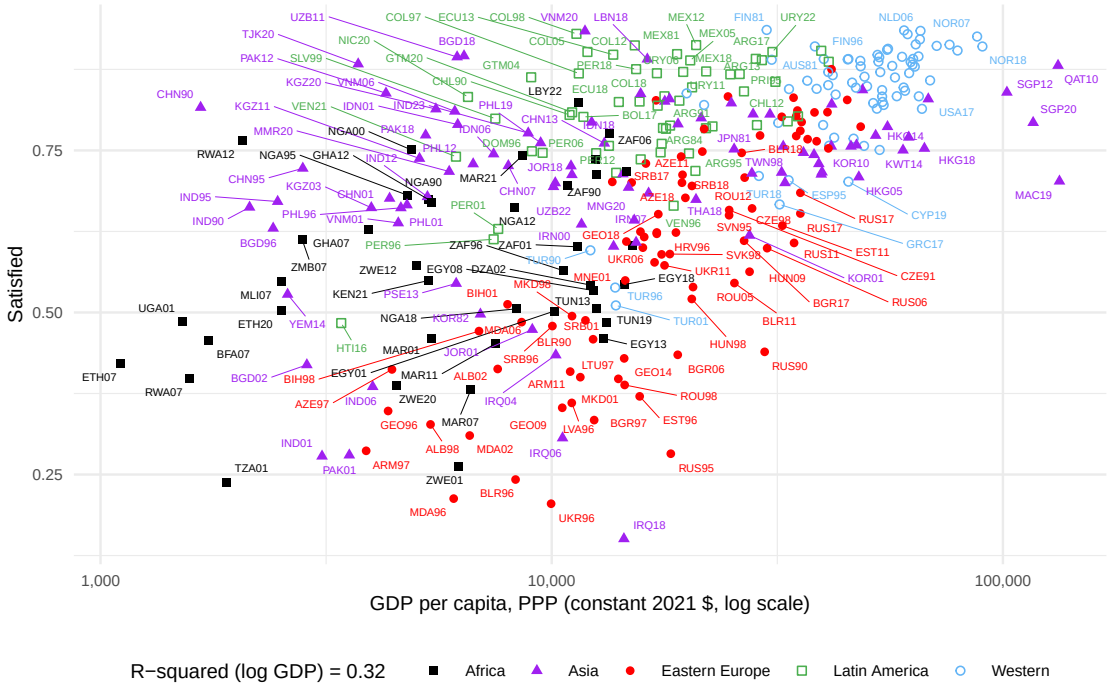


Figure A5: *Satisfied* vs. GDP per capita (PPP), WVS/EVS 1981–2023.

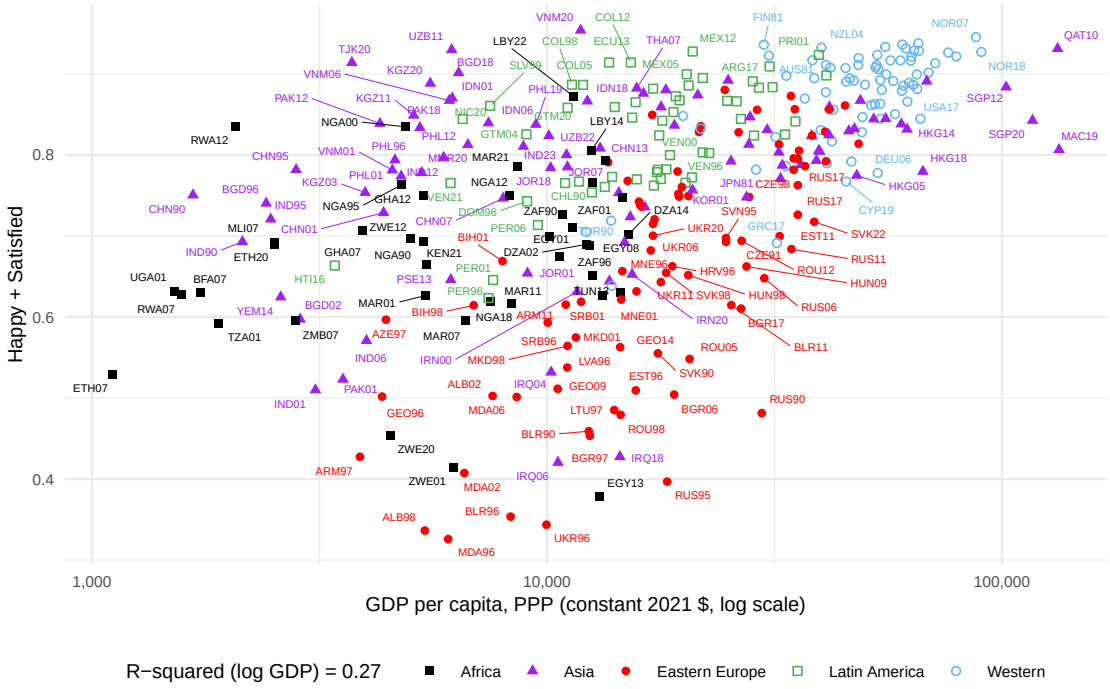


Figure A6: *Happy + Satisfied* vs. GDP per capita (PPP), WVS/EVS 1981–2023.

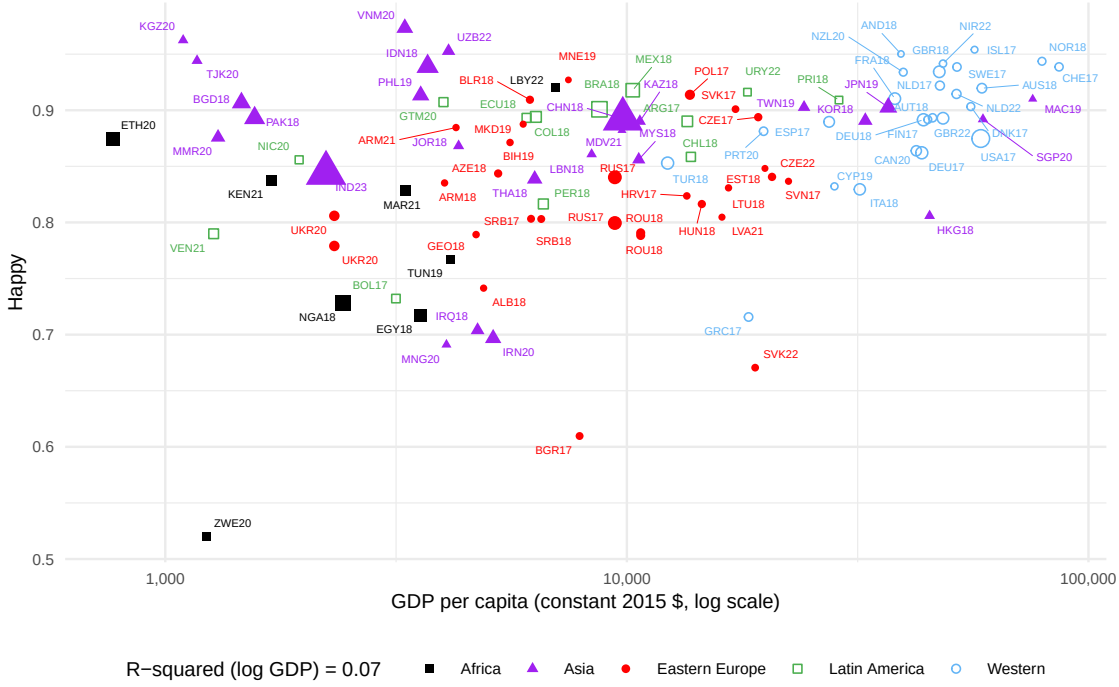


Figure A7: *Happy* vs. GDP per capita (nominal), WVS/EVS wave 7 (2017–2023), weighted by population.

Note: Point size and regression weights proportional to population. (Back to Section 3.2.)

References

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